



DEFENSE THREAT REDUCTION AGENCY
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FORT BELVOIR, VA 22060-6201

JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION

1. Nature and/or Description of Action

The Defense Threat Reduction Agency (DTRA) contracting activity proposes to procure, without using full and open competition, an effort entitled "X-ray Simulators for Test and Evaluation of Nuclear Survivability" as a new contract in support of the Department of Defense (DoD) Test Resource Management Center (TRMC) Central Test and Evaluation Investment Program (CTEIP).

2. Description of the Supplies/Services Required

DTRA requires support for the upgrade of the DTRA-owned Double-EAGLE (DE) and PITHON simulators, and related equipment and diagnostics located at the L-3 Applied Technologies Inc. (L-3) facility in San Leandro, CA. In addition, DTRA requires support for the upgrade of the Gamble II simulator capabilities at the Naval Research Laboratory (NRL) in Washington, DC. This support will require unique scientific, engineering and technical expertise developed over many years of DTRA-sponsored pulsed power Research and Development (R&D). The results of recent Fast Marx technology development for the Short Pulse Gamma simulator will be needed to conduct risk mitigation research and hardware design/development for upgrades on and to the simulators. The scope of this requirement includes activities to:

- a. Apply lessons learned from the Fast Marx development and demonstration to conduct and support simulator design studies and risk mitigation experiments to upgrade the pulsed power capabilities needed for enhanced cold (energies < 15 keV) and warm (15-100 keV) x-ray test environments on Gamble II, PITHON, and DE.
- b. Upon approval of the upgraded designs, fabricate, assemble, and demonstrate new pulsed power components and support systems at the existing L-3 and NRL facilities. The new capabilities should be designed to allow disassembly and relocation of the major components.

It is anticipated a sole-source contract with options will be awarded to L-3 for this effort. The CLINs for design, fabrication, development, assembly and demonstration are expected to be Cost Plus Fixed Fee. The CLIN for purchasing long lead items for capacitors are expected to be Firm Fixed Price. The program ceiling, to include all options under the contract, is estimated at \$80,000,000 over a period of five (5) years.

3. Statutory Authority

The statutory authority for this action is 10 United States Code (U.S.C.) 2304(c)(1) or 41 U.S.C. 253(c)(1), as implemented by Federal Acquisition Regulation 6.302-1, Only One Responsible Source and no other services will satisfy agency requirements.

4. Applicability of Authority

Only one source is capable of performing due to the unique or specialized nature of the work as follows:

L-3 is the only responsible source capable of providing the technical support needed for this requirement to be successful. For over forty years, L-3 has designed and built Nuclear Weapon Effects (NWE) simulators including the machines listed in paragraph 2 above. The DTRA-owned simulators built by L-3 and located at their San Leandro, CA facility, and the Gamble II at NRL are the only remaining high-level prompt cold and warm x-ray NWE testing and R&D capabilities outside of the National Nuclear Security Administration (NNSA) nuclear weapons laboratories.

L-3 has maintained and operated the DTRA-owned simulators, while providing a wide spectrum of radiation sources and diagnostics for effects testing by numerous government acquisition programs. L-3 is the designer and developer of the NWE simulators at their facility and of Gamble II at NRL. L-3 is also the developer of the radiation sources that will be operated and upgraded under the proposed contract. Furthermore, the upgraded DTRA and NRL owned simulators must remain in-place as the expertise, infrastructure, and business models needed to assure their continued availability to mission critical systems under development are fully established at the existing L-3 and NRL facilities.

The continued operation and upgrade of the simulators at the L-3 facility in San Leandro, CA, and NRL is currently the most efficient means of meeting DoD test support needs. Until NWE test support requirements increase to the point that the L-3 and NRL facilities cannot meet the test scheduling needs, it is not cost effective for DoD to develop or sponsor competitive contractor facilities.

5. Effort to Solicit Potential Sources

For this requirement, on October 10, 2018, DTRA issued a Sources Sought Announcement (ACRRDNTS181407) for R&D, testing, and engineering support for the potential X-Ray Simulators for Test and Evaluation of Nuclear Survivability (XSTENS) project. The purpose of the announcement was to identify firms capable of supporting the XSTENS project through three phases. Only one company responded, the incumbent, L-3 Applied Technologies Inc., San Leandro, CA.

For a previous contract, on February 6, 2012, DTRA issued a Sources Sought Announcement for R&D, testing and engineering support for the DTRA-owned NWE simulators and the L-3 Pulserad 1150 and sustainment of GFE located at the L-3 facility in San Leandro, CA. The purpose of the announcement was to identify firms capable of providing the required services to support the simulators. Only one company responded to that announcement as well, the incumbent, L-3 Applied Technologies Inc., San Leandro, CA. Additionally, there was a review of past open announcements for sources sought that were conducted for Contract Number HDTRA2-08-C-0002 that resulted in no viable alternative providers being identified.

6. Fair and Reasonable Costs

The anticipated cost/price to the Government of this contract action will be determined to be fair and reasonable based on inputs and evaluations from the DTRA Program Office, analysis of cost and pricing data, field pricing support from the Defense Contract Audit Agency (DCAA) and/or Defense Contract Management Agency (DCMA), and historical data.

7. Market Research

The market research concluded that L-3 is the only responsible source that can meet the requirements of this effort. The project manager used the following techniques to conduct market research:

- a. Historical acquisition information (review of market research results for similar or identical services), to include the results of a sources sought research conducted for the current Contract Number HDTRA2-08-C-0002 that resulted in no viable alternative contractor;
- b. Reference materials such as the Guide to Nuclear Weapon Effects Simulation Facilities and Techniques (2014 update);
- c. A broad Internet search for pulsed radiation sources and a specific search of listings in the Thomas Register (<http://www.thomasnet.com/products/radiation-sources-76510502-1.html>);
- d. Publication of a Sources Sought Announcement in FedBizOpps on October 10, 2018 to which only one company responded, L-3 Services, Inc., Applied Technologies, San Leandro, CA; and
- e. Personal knowledge in procuring services of this type, to include consultation with other branches in DTRA and NRL that are familiar with past contract activities, capabilities, and the expertise of personnel at L-3 and other contractors in the DoD and DOE nuclear survivability community.
- f. FAR Part 8 was reviewed for consideration of mandatory sources. This requirement is not on the services procurement list maintained by the Committee for Purchase From People Who are Blind or Severely Disabled and therefore not viable under the AbilityOne program. Furthermore, a review of non-mandatory sources was also considered. This requirement is not on the list of services provided by Federal Prison Industries, Inc. and thus not a viable option.

The market research confirmed that due to the highly specialized nature of the requirement, L-3 is the only responsible source that can satisfy the need. Factors that led to this conclusion are that: (1) L-3 has the scientific and engineering expertise in NWE and pulsed radiation source technology and (2) can successfully provide the required services. A new contractor would be required to duplicate L-3's unique knowledge and expertise, which would take years, and require resources above and beyond the level of effort by L-3. Even if trained, having an outside contractor operate the DTRA-owned simulators in the facility leased by L-3 would limit access to shared resources needed to support machine operations. L-3 has indicated that it could not allow another contractor to operate independently within its facility without raising significant issues of liability and safety. Relocating the upgraded PITHON and DE machines to another facility has been determined in both the recent Analysis of Alternatives (AoA) and two prior feasibility studies for the consolidation of DoD test capabilities to be prohibitively expensive and technically risky. See the market research report for additional details.

8. Other Facts

For another contractor to do the work, the technical drawings of the existing simulators and radiation sources would have to be updated by L-3 and provided to another contractor in addition to training new technical and scientific personnel on the details of machine, radiation source, and diagnostic operations. The additional training effort and duplication of expertise by another contractor is estimated to cost an additional \$6,000,000 to the Government and take a minimum of two years of joint effort to accomplish. This would not be a feasible or an acceptable approach to meet this critical requirement, nor is such funding available.

9. Interested Sources

Besides L-3, there were no other respondents who expressed written interest to the sources sought for this requirement.

10. Subsequent Actions

As long as nuclear weapons remain as threats to the U.S., critical military systems must be certified to be survivable to the environments they could produce. Both U.S. military and commercial sectors rely heavily on space assets, whose mission capabilities can be jeopardized by damage from prompt nuclear radiation. This need to support the certification of DoD systems adds urgency and importance to fulfilling this requirement in the most effective and expedient manner possible. However, based on numerous studies and projections of requirements for the level of NWE test capabilities embodied at the L-3 facility, it was determined that their facility could meet the needs of the DoD survivability community. Until NWE requirements increase to the point that the L-3 facility cannot meet the test scheduling needs, it is not cost effective for DoD to fund the development of alternative or new simulator capabilities. However, DTRA will continue to survey the community and monitor the market to research the development or maturity of a comparable system in order to assess the potential for competition in the future for work related to this requirement.

Technical Certification

I certify that the data and information forming the basis for this justification are accurate and complete to the best of my knowledge and belief.



Contracting Officer's Representative

Contracting Officer Certification

I certify that this justification is accurate and complete to the best of my knowledge and belief.

CHANG.DAISY.M.1264301919 Digitally signed by CHANG.DAISY.M.1264301919
Date: 2019.04.24 06:53:16 -04'00'

Daisy M. Chang
Contracting Officer

Date

General Counsel Coordination

This justification is legally sufficient.



Associate General Counsel

Approval

Based on the above justification, I hereby approve accomplishment of this acquisition by means other than using full and open competition.



Head of Contracting Activity

Date